MIL-C-52774A(ME)
8 October 1976
SUPERSEDING
MIL-C-52774(ME)
2 January 1974

MILITARY SPECIFICATION

CRADLE, 27 FOOT BRIDGE ERECTION BOAT

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cradle designed to transport, launch, and retrieve the standard 27 foot bridge erection boat from a ribbon bridge transporter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

FSC 5420

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1261	- Welding Procedures for Construction Steels.

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E1010	- Cradle, 27 Foot Bridge Erection Boat.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.5 - Low-Alloy Steel Covered Arc-Welding Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The cradle shall be in accordance with Top Assembly TA13220E1010 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the requirements shown on the drawing is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more cradles as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce cradles that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified, the Government will conduct any or all of the preproduction examinations and tests as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more cradles for inspection and testing specified in 4.6.1 and 4.6.2.

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3.4 Material. Materials shall be as specified herein and on the applicable drawings. The surfaces of materials shall be free of foreign matter, scale, rust, or corrosion.

3.4.1 Welding electrodes. Unless otherwise specified on the end product drawings, the following welding electrodes shall be used.

3.4.1.1 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, Classification ER 5356.

3.4.1.2 For steel. Welding electrodes for steel shall be as specified in MIL-STD-1261 except that electrodes conforming to AWS A5.5, Classification E11015, E11016, E11018, or E11018-M shall be used for high strength steels.

3.5 Reliability, Availability, Maintainability (RAM).

3.5.1 Reliability. The cradle shall have a 0.98 probability of a mission success when tested as specified in 4.6.2.1. A mission is defined as one launch and retrieval cycle and 20 miles of transport (50 percent paved road and 50 percent gravel road).

3.6 Lubrication. Lubrication shall be as shown on the end product drawings.

3.7 Cleaning, treatment, and painting. Cleaning, treatment, and painting shall be shown on the end product drawings.

3.8 Identification marking. The cradle and components shall be identified as shown on the drawings.

3.9 Performance. The cradle dolly shall roll freely on the rails from one end of the cradle to the other. The release latch shall hold the dolly securely in position when engaged, and shall release the dolly without forcing when the release rope is pulled. The cradle guide roller shall roll freely. All pins shall fit in their intended holes without forcing.

3.10 Government-loaned property. When specified (see 6.2), a Ribbon Bridge Transporter (MIL-T-52735), will be loaned by the Government (see 6.6).

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3.11 Workmanship.

3.11.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing for steel only. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges are required or where they are not detrimental to safety.

3.11.2 Welding. Welding shall be in accordance with the specifications and standards referenced on the drawings. Aluminum welding shall be done by the gas-metal-arc (MIG) method.

3.11.3 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on the applicable drawings, and to the standards of the industry.

3.11.4 Jigs and fixtures. Components (except minor parts) shall be assembled in jigs or fixtures and joined while held in position. Jigs or fixtures shall be designed to minimize distortion of the parts being welded.

3.11.5 Interchangeability. All parts, components, and assemblies having the same identification number shall be functionally and dimensionally interchangeable. Interchangeability is defined as two or more parts (components and assemblies) possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of themselves or of adjoining parts (components and assemblies) except for built-in adjustment and without selection for fit and performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.2 Parts and components. Parts and components detailed on the end product drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan and the basis for acceptance and rejection (see 6.7 and 6.8.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction cradle(s) or separately procured component(s) shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction cradle shall be tested as specified in 4.6.2.1. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production cradles will be selected at random by the Government from the cradles being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production cradle shall not exclude the remaining cradles from the quality conformance inspection and acceptance provisions specified in Section 4.

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4.4.1 Inspection failure. Failure of an initial production cradle to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production cradle and shall be cause for refusal by the Government to continue acceptance of production cradles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cradles previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all cradles accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination for separately procured cradle components shall be in accordance with MIL-STD-105, Inspection Level II.

4.5.2 Examination.

4.5.2.1 Individual. Each cradle shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2.2 Samples. Samples of separately procured components selected in accordance with 4.5.1 shall be examined for the applicable defects specified in 4.6.1. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The boat cradle(s) shall be examined as specified herein for the following defects:

101. Any part or component not in accordance with SQAP requirements as shown on the end product drawings.
102. Any dimension other than those identified on the SQAP requirements shown on the drawings not as specified.
103. Treatment and painting not as specified.
104. Members not true to dimension and line, containing twists, bends or other imperfections.

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105. Weld defects, such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion or spatter, or incorrect size.
106. Welding not performed in accordance with qualified procedure.
107. Material not as specified.
108. Lubrication and lubrication fittings not as specified on the drawings.
109. Commercial components not as specified.
110. Components not interchangeable with respect to installation, performance, and quality.
111. Nonskid compound not applied correctly or missing.
112. Identification marking missing or incorrect.
113. Components missing or not as specified.
114. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Operational. With the cradle aboard a Ribbon Bridge Transporter with a 27-Foot Bridge Erection Boat in place as specified in TM 5-2090-200-12, 10 launchings and retrievals of the boat shall be accomplished. After the 10th retrieval the boat shall be secured and transported over various types of road surfaces (50 percent paved and 50 percent gravel) for 200 miles. This cycle shall be repeated until a total of 100 launches and retrievals and 2000 miles of transporting have been completed. This test will be considered equivalent to 100 missions. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.2 Reliability. Using the results of the operational test (4.6.2.1), reliability failures will be assigned in accordance with 4.6.2.2.1 The occurance of three or more failures shall constitute failure of the reliability requirement 3.5.1.

4.6.2.2.1 Failure definition. A failure is defined as any malfunction which cannot be corrected within 15 minutes by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

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(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
(b) Damage to the cradle by continued operation.
(c) Safety hazards to personnel.

4.7 Inspection comparison. The Government may select cradles at any time during the contract production period and subject these cradles to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on cradles selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production cradles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison cradle to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison cradle and shall be cause for refusal by the Government to continue acceptance of production cradles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cradles previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all cradles accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Examination. Each cradle shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

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115. Disassembly not as specified.
116. Preservation not in accordance with referenced document for Level A.
117. Consolidation not as specified for Level A.
118. Consolidated boxes and tubular uprights not strapped to cradle frame as specified for Level A.
119. Marking missing, illegible, or incorrect.

5. PACKAGING

5.1 Disassembly. The tubular uprights, turnbuckles, clevises, ropes and associated hardware shall be disassembled from the cradle. Attaching hardware shall be secured in one of the mating parts.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Technical publication. Technical publications shall be preserved Method IC-1 or MIL-P-116.

5.2.1.2 Preservatives. Items requiring the application of a preservative shall be preserved in accordance with the applicable criteria of MIL-P-116.

5.2.1.3 Consolidation. The repair parts, maintenance tools, turnbuckles clevises, ropes and associated hardware shall be consolidated in boxes conforming to PPP-B-601, Domestic Type, Grade B or PPP-B-621, Class 1.

5.2.2 Commercial. The repair parts, maintenance tools, turnbuckles, clevises, ropes, technical publications and associated hardware shall be placed in fiberboard or wooden boxes which will afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A. The cradle dolly shall be fully retracted in the cradle and secured. The consolidated boxes and tubular uprights shall be strapped to

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the cradle frame in a manner which will not increase cubage or interfere with lifting. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish B.

5.3.2 Commercial. The cradle dolly shall be fully retracted in the cradle and secured. The cradle and associated equipment shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The boat cradle is intended to be used to transport the standard 27 foot bridge erection boat on a Ribbon Bridge Transporter and to launch and retrieve the boat.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of cradles (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examinations. When the Government elects to conduct a portion of the preproduction examination, the contracting officer should specify which examination will be conducted by the Government and which examination shall be conducted by the contractor.
(d) When initial production inspection is required and number of cradles to be furnished (see 3.3).
(e) Degree of preservation and packing required (see 5.2 and 5.3).

6.3 Preproduction model. Any changes or deviations of production cradles from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish cradles conforming to this specification.

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6.4 Data requirements. The contracting officer should inlcude requirements for such data as technical publications, instructional materials, illustrated parts lists and contractors maintenance and operational manual to be furnished with each cradle.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of cradles.

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified (see 3.10).

6.7 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.8 Definitions.

6.8.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of this specification. SQAP indicates the minimum requirements which must be inspected on the end product drawings to verify the design objective of the product and assure interchangeability of repair parts.

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Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE
MIL-C-52774A(ME) Cradle, 27 Foot Bridge Erection Boat

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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